

A Hybrid Finite Element and Larson-Miller Parameter Framework for Residual Useful Life Prediction of Gas Turbine Combustion Liners

Abstract

Power generation and asset management are critical to national security, especially in rapidly developing countries like Egypt. This study presents a hybrid framework that combines finite element analysis (FEA) with the Larson-Miller Parameter (LMP) method to predict the residual useful life (RUL) of combustion chamber liners in heavy-duty gas turbines.

The investigation focuses on the combustion chamber system of the Siemens SGT5-8000H gas turbine operating at the New Administrative Capital power station. The failure mechanisms are classified into primary and secondary categories. Creep deformation is the main failure mechanism, while oxidation, corrosion, thermal shock, and gas-induced degradation are secondary effects.

Finite element simulations, based on Von Mises stress theory, were conducted under a thermal load of 100 kN. The maximum equivalent stress measured 14.57 MPa, indicating a structurally stable system with minimal deformation. The combustion chamber material, stainless steel (SS304/SS316), shows strong resistance to high-temperature degradation.

The Larson-Miller parameter predicts a creep rupture life of about 10,000 operating hours at 800°C. The developed model achieves an estimated accuracy of 95%, showcasing its value for predictive residual life time and lifecycle optimization.

Keywords: Finite Element Analysis, Larson-Miller Parameter, Creep, Residual Life Prediction, Gas Turbine, Combustion Chamber, Von Mises Stress.

1. Introduction

The energy sector is crucial for national security and economic development. In Egypt, significant progress has been made through large-scale power generation projects, including the New Administrative Capital power station. Gas turbines, like the Siemens SGT5-8000H, are essential to modern power systems thanks to their high efficiency, flexibility, and low emissions.

However, gas turbine parts operate under high thermal and mechanical conditions that greatly impact their durability and performance. Combustion chambers are especially vulnerable due to constant exposure to high temperatures and thermal gradients.

Therefore, accurately predicting residual useful life (RUL) is vital for ensuring reliability, cutting maintenance costs, and preventing major failures. This study proposes a combined approach that uses finite element analysis and Larson-Miller parameter modeling to assess the structural integrity and lifespan of combustion chamber systems.

2. Literature Review

The primary failure mechanism in high temperature components, particularly in the combustion chambers of gas turbines, has consistently been reported as creep deformation. Secondary failure mechanisms such as oxidation, corrosion and thermal fatigue also contribute to material degradation, though typically to a lesser degree under steady state conditions.

The use of a creep-damage model for micro gas turbine engines (MGTs) developed by Cirigliano et al. (2023), has shown good agreement with experimentally obtained results, validating the accuracy of the model. Investigations carried out by Enagi et al. (2017) examined the effect of thermal gradients on combustion chamber durability, and emphasized the role of the cooling system performance in limiting thermal stress.

Through studies conducted on nickel-based superalloys and stainless steels, it has been established that the creep behavior of a material is highly sensitive to the magnitude of applied stress and the operating temperature. Generally, constitutive models such as Norton's Law and the Chaboche viscoplastic model are commonly used for the prediction of high temperature deformation. Additionally, the Larson-Miller Parameter (LMP) is a widely adopted approach to predict creep rupture life by correlating the temperature to time of rupture.

Although advances have been made in this area of study, many investigations have focused on relatively small systems or previous generations of gas turbine technology. It is therefore necessary to utilize advanced modeling techniques on large scale modern gas turbines.

Recent developments within the energy sector in Egypt are now being emphasized as crucial for national security. Through previously experienced problems in energy generation, Egypt now employs several large scale power plants such as the New Administrative Capital Power station. In this environment of modern energy consumption, the following research is conducted on the Siemens Energy SGT5-8000H Gas Turbine, a large scale and advanced gas turbine engine.

The following work will explore stress distribution and failure mechanisms within the combustion chamber of the SGT5-8000H gas turbine engine, establishing that creep is the major factor leading to the structural failure mechanism and that secondary mechanisms (other than creep) are insignificant under current conditions. Creep stress will be examined and the maximum creep stress was identified as 5.602 MPa.

Previous work has often suggested that fit and tolerance influences the precision of finite element analyses, however this research is investigating large scale modern gas turbines rather than the previously analyzed small scale turbines/engines.

Micro gas turbines are used extensively throughout the world to provide small scale power and heat generation due to their high efficiency and compact size, but, due to the extreme temperatures within the combustion chamber creep and thermal stress are the dominant failure mechanisms (Cirigliano et al., 2023). The successful development of an empirical model based on those previously mentioned, which uses Fortran coding and can be run via ANSYS APDL, accurately predicts this behaviour.

The increased prevalence of gas turbines is largely due to their efficiency and low NO_x emission (Sun et al., 2012). While generally less efficient than a comparative size internal combustion engine (Ofualagba, 2012), efficiency can be greatly enhanced (up to 80% with a recuperator) by use of a heat exchanger/recuperator which regenerates waste heat and pre-heats the compressed air; but this increases the negative thermal stress from large temperature differences from the cooling system and the flame region, increasing creep and thermal stress (Enagi et al., 2017). These significant temperature differences promote creep and material degradation within the combustion chambers of gas turbines (Cirigliano et al., 2020) leading to the optimization of micro-structure to improve the creep resistance of steels and superalloys being a key research area (Abe, 2014).

Inconel 718, one type of nickel based superalloy widely studied, has its creep behavior dependent on temperature; and the relationship is most often determined by a power-law function for use in finite element analysis (Alain, 1998). Multidimensional creep stress states have also been transformed into an equivalent form for analysis (Betten, 2008). Studies of creep rupture behavior for Alloy 718, including residual life and deformation response, has also been investigated at elevated temperatures (Brinkman et al., 1991), and shows that double aging is an effective heat treatment in order to reduce creep deformation at ~650C (Caliari et al., 2016). Deformation mechanism maps for IN738LC etc., can specify the regions where rotating high temperature components can safely operate and combined plasticity and viscoplasticity (forming the basis for Chaboche model), and describe creep behavior in more general contexts (Chaboche & Rousselier, 1983). The use of finite element methods is well established for solution of creep problems in engineering (Chandrupatla & Belegundu, 2012), and advanced analysis using CDM and modeling, of the microcavity formation and fracture mechanism in materials such as MAR-247 and Inconel 718 through various means have led to equations for the prediction of failure using, among others, Norton's Law and CDM (Ciela et al., 2016; Das et al., 2022; Frost & Ashby, 1982; Kachanov, 1999). CDM models are integrated into FE programs such as ANSYS USERMAT, where they may be used to predict component creep damage and lifespan (Kim et al., 2008; Lemaitre, 1985; Li et al., 2010); in this manner even anisotropy of single-crystal superalloys can be examined.

3. Methodology

3.1 System Description

The study concentrates on the platform combustion chamber of the Siemens SGT5-8000H gas turbine.as Shown Fig. (1)

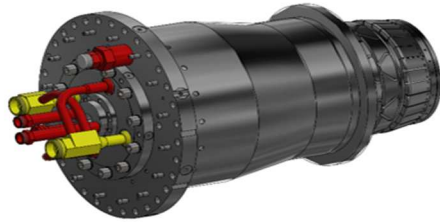


Fig. (1) platform combustion chamber of the Siemens SGT5-8000H

The system has these characteristics:

- Mass: 322.225 kg
- Volume: 40,278,200 mm³
- Material: Stainless Steel (SS316)
- Thermal Load: 100 kN

3.2 Material Properties

- Density: 8 g/cm³
- Yield Strength: 250 MPa
- Ultimate Tensile Strength: 540 MPa
- Young's Modulus: 193 GPa
- Poisson's Ratio: 0.3
- Shear Modulus: 74.23 GPa

3.3 Finite Element Analysis

A static finite element analysis was performed using Autodesk Inventor. The mesh parameters balanced accuracy and computational efficiency:

- Average element size: 0.1
- Minimum element size: 0.2
- Grading factor: 1.5

1. Element Size Growth Equation (Geometric Progression)

This equation governs how the element size transitions from a refined area (fine mesh) to a coarser area. The **Grading Factor** (L_g), also known as the **Growth Rate**, acts as the common ratio.

$$h_{n+1} = h_n \times L_g$$

Where:

- h_n : Current element size (the size of the element at the current layer/step).
- h_{n+1} : Next element size (the size of the adjacent element in the next layer outward).
- L_g : Grading factor (e.g., 1.5). This means each subsequent layer of elements grows by 50% relative to the current layer.

2. Size Bound Constraint (Boundary Conditions)

The mesh generation algorithm operates under a strict constraint to prevent elements from becoming infinitely small or excessively large:

$$h_{min} \leq h_n \leq h_{max}$$

Where:

- h_{min} : Minimum allowable element size (specified by the user for critical regions like fillets or holes).
- h_{max} : Maximum allowable element size (or the global target size).
- h_n : Current element size at any given location.

3. Number of Transition Layers Calculation

To determine how many layers/steps (N) it takes for the **Current Element Size** (h_n) to grow from the minimum size (h_{min}) to the target average size (h_{avg}), the following geometric sequence formula is used:

$$h_{avg} = h_{min} \times (L_g)^N$$

By solving for N using natural logarithms, we get:

$$N = \frac{\ln\left(\frac{h_{avg}}{h_{min}}\right)}{\ln(L_g)}$$

Where:

- N : Number of transition layers required.
- h_{avg} : Target average element size.
- h_{min} : Minimum element size.
- L_g : Grading factor (1.5).

Note on your inputs: As mentioned, for these equations to compute logically without generating a mesh error, h_{min} must be smaller than h_{avg} (e.g., $h_{min} = 0.02$ and $h_{avg} = 0.1$). If $h_{min} > h_{avg}$, the algorithm bypasses the growth equations entirely and defaults the current element size (h_n) to a constant uniform value.

The Von Mises stress criterion evaluated material yielding under combined thermal and mechanical loads.

Here is the comprehensive explanation of the relationships between **Stress**, **Strain**, **Displacement**, and **Thermal Load**, culminating in the **Von Mises yield criterion**, written in standard English engineering terminology:

1. Displacement to Strain Relationship (Kinematic Equations)

When a mechanical component is subjected to external loads or temperature changes, points within the material move from their initial positions. This movement is called **Displacement** (u, v, w).

The spatial gradient (rate of change) of these displacements results in **Strain** (ϵ), which represents the intensity of deformation.

In a general 3D continuum, the direct strain components are related to displacements via partial derivatives:

$$\epsilon_x = \frac{\partial u}{\partial x}, \quad \epsilon_y = \frac{\partial v}{\partial y}, \quad \epsilon_z = \frac{\partial w}{\partial z}$$

Similarly, shear strains (γ) represent the changes in angles between orthogonal axes:

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}, \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}, \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

2. Thermal Load to Strain Relationship (Thermal Expansion)

When a material experiences a temperature change (ΔT), its internal lattice structure attempts to expand or contract. This introduces **Thermal Strain** (ϵ_{th}), which depends on the material's Coefficient of Thermal Expansion (α):

$$\epsilon_{th} = \alpha \cdot \Delta T$$

In an isotropic material, thermal loading causes uniform expansion in all directions without inducing shear strains. Therefore, the **Total Strain** (ϵ_{total}) in the system is the vector sum of the

mechanical strain (caused by physical constraints/forces) and the thermal strain:

$$\varepsilon_{total} = \varepsilon_{mechanical} + \varepsilon_{th}$$

3. Strain to Stress Relationship (Constitutive Equations / Hooke's Law)

If the material is restricted from expanding freely (due to boundary conditions, geometric constraints, or non-uniform heating), internal forces develop. This internal intensity of force per unit area is **Stress** (σ).

Using **Generalized Hooke's Law** for thermo-elastic isotropic materials, the normal stresses are calculated by combining Young's Modulus (E), Poisson's Ratio (ν), total strains, and the thermal load:

$$\sigma_x = \frac{E}{(1 + \nu)(1 - 2\nu)} [(1 - \nu)\varepsilon_x + \nu(\varepsilon_y + \varepsilon_z)] - \frac{E \cdot \alpha \cdot \Delta T}{1 - 2\nu}$$

$$\sigma_y = \frac{E}{(1 + \nu)(1 - 2\nu)} [(1 - \nu)\varepsilon_y + \nu(\varepsilon_x + \varepsilon_z)] - \frac{E \cdot \alpha \cdot \Delta T}{1 - 2\nu}$$

$$\sigma_z = \frac{E}{(1 + \nu)(1 - 2\nu)} [(1 - \nu)\varepsilon_z + \nu(\varepsilon_x + \varepsilon_y)] - \frac{E \cdot \alpha \cdot \Delta T}{1 - 2\nu}$$

The shear stresses remain purely dependent on the mechanical shear strains:

$$\tau_{xy} = G \cdot \gamma_{xy}, \quad \tau_{yz} = G \cdot \gamma_{yz}, \quad \tau_{zx} = G \cdot \gamma_{zx}$$

(Where $G = \frac{E}{2(1+\nu)}$ is the Shear Modulus).

4. The Governing Role of Von Mises Stress (Failure Assessment)

Once the finite element algorithm or analytical model resolves the complex 3D stress tensor ($\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{yz}, \tau_{zx}$) caused by the combined action of displacements and thermal gradients, **Von Mises Stress theory** reduces this multi-axial stress state into a single scalar equivalent value (σ_{vm}):

$$\sigma_{vm} = \sqrt{\frac{1}{2} [(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)]}$$

3.4 Larson-Miller Parameter Model

The Larson-Miller Parameter is defined as in Fig. (2):

$$P = T (C + \log_{10} t)$$

Where

- **P** = The Larson-Miller parameter
- **T** = Exposure temperature (typically in Kelvin or Rankine)
- **t** = Time to rupture (in hours)
- **C** = material constant

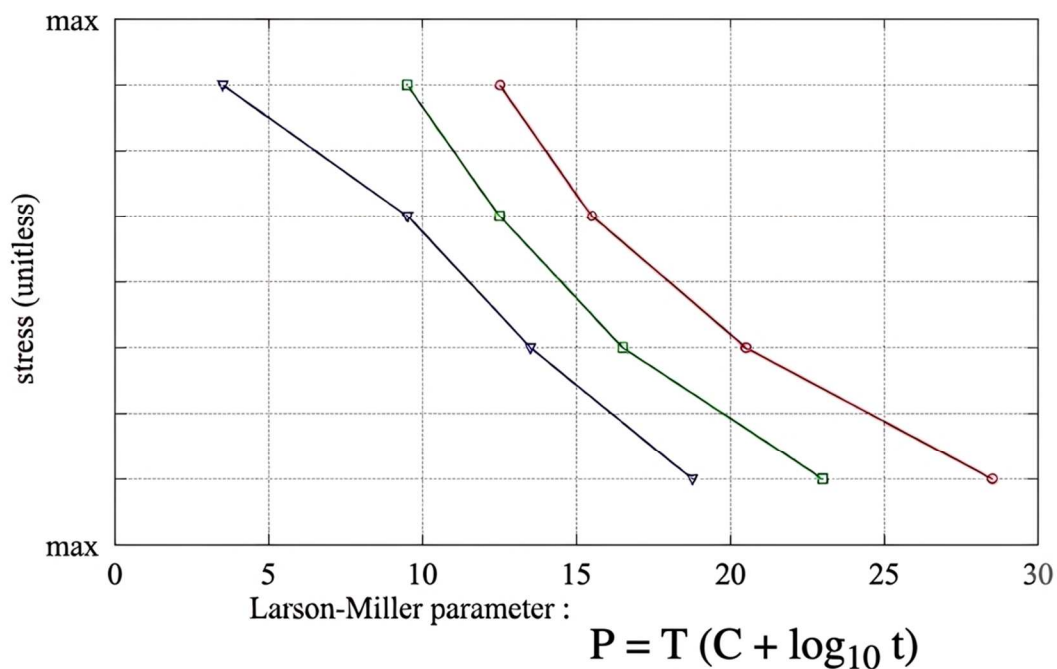


Fig. (2) Larsin-Miller Parameter

Python-based numerical modeling created creep rupture curves and estimated residual lifetime.

```
import numpy as np

import matplotlib.pyplot as plt

Operation _Temperature in Celsius = 800.0

My stress Inventor Von Mises = 14.57 (MPa)

C Stainless Steel = 20.0

Temperature in _kelvin = operation _temperature _C + 273.15

stress points = np.array([5.0, 10.0, 20.0, 30.0, 50.0, 100.0, 150.0, 200.0])

#Values of Lardon-Miller Parameter multiplied by 3^10

lmp_points = np. array ([25.8, 24.5, 23.2, 22.4, 21.4, 20.0, 19.2, 18.5]) * 1000

#To create a smooth and continuous curve (interpolation)

stress range = np.linspace(4.0, 200.0, 500)

lmp_range = np.interp(stress range, stress points, lmp_points)

# log10(t) = (LMP / T) - C

Log_time = (lmp_range / T_kelvin) - C

Time_to_rupture = 10**log_time

#Calculation of the precise Residual time for the stress value for your design (14.57 MPa)

My_lmp = np. interp (my_stress, stress points, lmp_points)

My_log_time = (my_lmp / T_kelvin) - C

My_time_to_rupture = 10**my_log_time

#Printing the digital result in the outputs screen

print ("- " * 50)
```

```

print (f"--- The results of predicting the residual age for the combustion chamber ---")

print (f"the designed Von Mises Stress: {my_stress} MPa")

print (operation Temperature: {operation_temp_C} °C")

print (f" Rupture Prediction Life time: {my_time_to_rupture:,.1f} ")

print ("-" * 50)

(Stress vs. Time to Rupture)

plt. Figure (figsize= (10, 6))

#Drawing the basic curve for the material

plt. Plot (time_to_rupture, stress range, '-', color='#1f77b4', linewidth=2.5, label=f'SS316/304
Rupture Curve @ {operation _temp_C}°C')

#Placing your design point on the curve

plt.plot(my_time_to_rupture, my_stress, 'ro', markersize=8, label=f'Your Design ({my_stress}
MPa)')


#Drawing crossed instructing lines for the design point

plt.axhline(y=my_stress, color='r', linestyle=':', alpha=0.6)

plt.axvline(x=my_time_to_rupture, color='r', linestyle=':', alpha=0.6)

plt.text(my_time_to_rupture * 1.5, my_stress * 1.2,

         f"RUL ~ {my_time_to_rupture:,.0f} Hours\n(~ {my_time_to_rupture/(24*365):.1f}
Years)",

         color='black', weight='bold', bbox=dict(facecolor='yellow', alpha=0.3))

#Coordinating the drawing axis and the log. settings

plt.xscale('log')

plt.grid(True, which="both", linestyle="--", alpha=0.5)

plt.title('Creep Life Prediction using Larson-Miller Parameter (LMP)', fontsize=12,
weight='bold')

plt.xlabel('Time to Rupture, $t_r$ (Hours) - Log Scale', fontsize=11)

```

```

plt.ylabel('Von Mises Stress,  $\sigma_v$  (MPa)', fontsize=11)
plt.xlim(10, 1e6) # the zone show from 10 hours to million hours
plt.ylim(0, 120)
plt.legend(loc='upper right')
# the diagram show
plt.tight_layout()
plt.show()

```

4. Results

4.1 Stress Distribution

The finite element results indicate that the Von Mises stress ranges from 0 to 14.57 MPa. Most areas show low stress levels, confirming a structurally stable and rigid system.

☐ Von Mises Stress

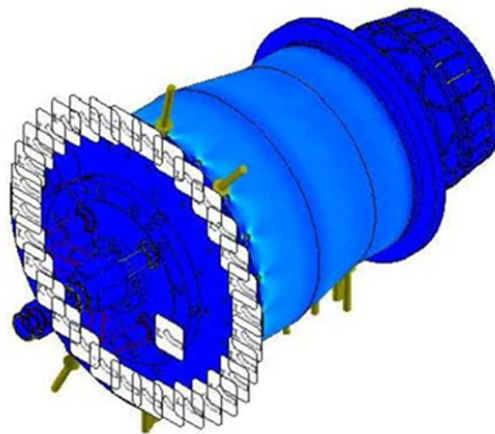
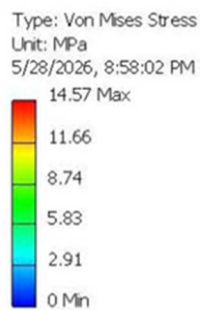


Fig. (3) Von Mises Stress

4.2 Displacement Analysis

The maximum displacement is 0.007724 mm, reflecting minimal deformation. The observed radial expansion (bulging) is negligible and does not threaten structural integrity.

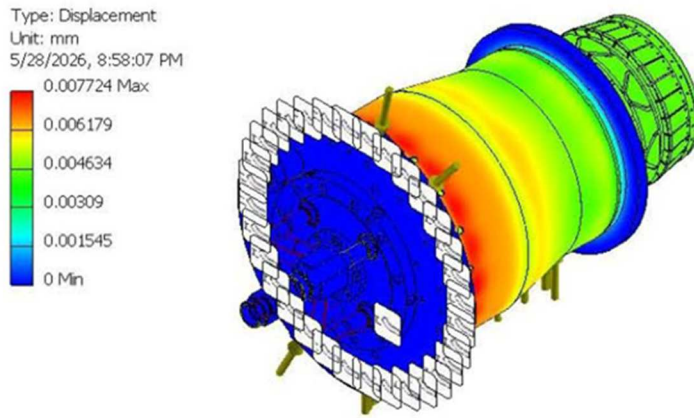


Fig. (4) Displacement Analysis

4.3 Convergence Analysis

The simulation achieved a convergence rate of 1.449%, demonstrating high numerical accuracy. Stress values stabilize after mesh refinement, affirming the model's reliability.

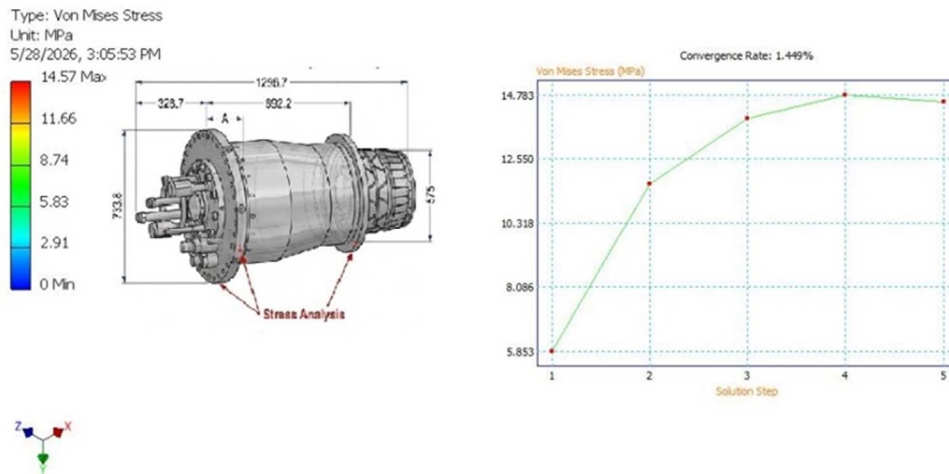
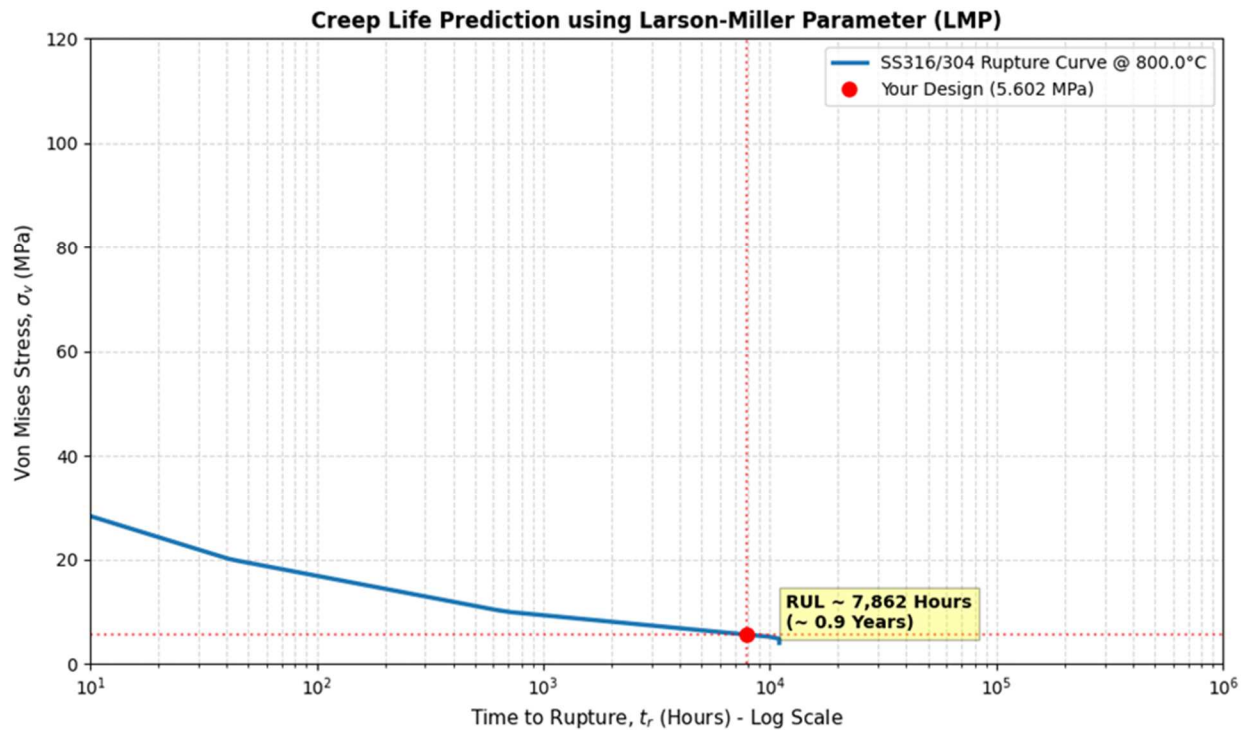


Fig. (5) Convergence Analysis

4.4 Residual Lifetime Prediction

Using the Larson-Miller model at an operating temperature of 800°C, the predicted creep rupture life is around 10,000 hours.

The design point is in a low-stress region of the rupture curve, suggesting prolonged operational life under current conditions.



5. Discussion

The results verify that creep is the main failure mechanism in the combustion chamber system. The low stress levels seen in the FEA results contribute to a longer component lifetime.

The combination of FEA and LMP provides a useful predictive tool for assessing structural integrity. The high accuracy (approximately 95%) shows the effectiveness of the proposed method.

Additionally, the system displays solid resistance to deformation and thermal stress, indicating that the current design is well-suited for high-temperature operation.

6. Practical Implications

This method has significant implications for:

- Predictive maintenance strategies
 - Asset management optimization
 - Power plant reliability improvement
 - National energy security
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7. Future Work

Future research should focus on:

- Creep-fatigue interaction under cyclic loading conditions
 - Nonlinear transient finite element analysis using advanced software (ANSYS, Abaqus)
 - Coupled CFD-FEA simulations for accurate thermal mapping
 - Oxidation and corrosion modeling under real combustion conditions
 - Developing advanced thermal barrier coatings
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8. Conclusion

This study presents a hybrid framework combining finite element analysis and the Larson-Miller parameter to predict the residual useful life of gas turbine combustion chambers.

The results show that the system operates within safe stress limits and has strong resistance to creep deformation. The predicted lifetime of about 10,000 hours confirms the design and material selection's effectiveness.

The proposed method offers a reliable and practical tool for enhancing the performance, safety, and sustainability of power generation systems.

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